

Exp No:3	Setting up Hybrid Cloud Connectivity using Cisco Packet Tracer
Date: 22/07/2026	

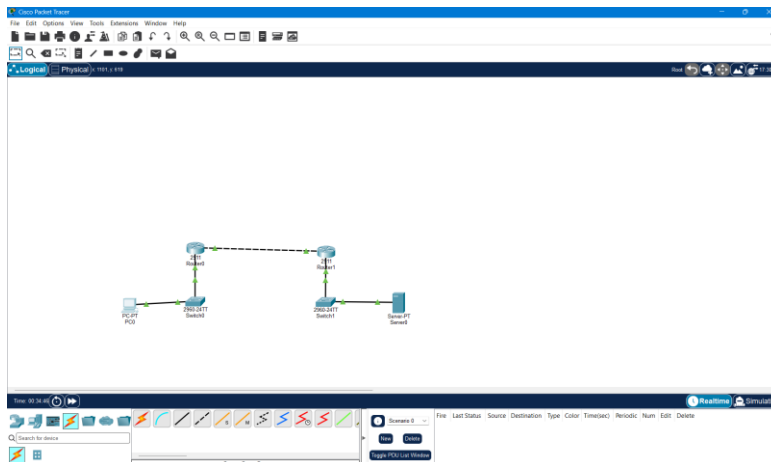
Aim:

To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer.

Software Used:

- Cisco Packet Tracer
- Windows 11 (or your OS)

Network Topology:



Hybrid cloud network topology.

- PC
- Switch0
- Router0

- Router1
- Switch1
- Server
- All green links

Router0 Configuration

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          down
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          up
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.255.255.0 10.10.10.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#

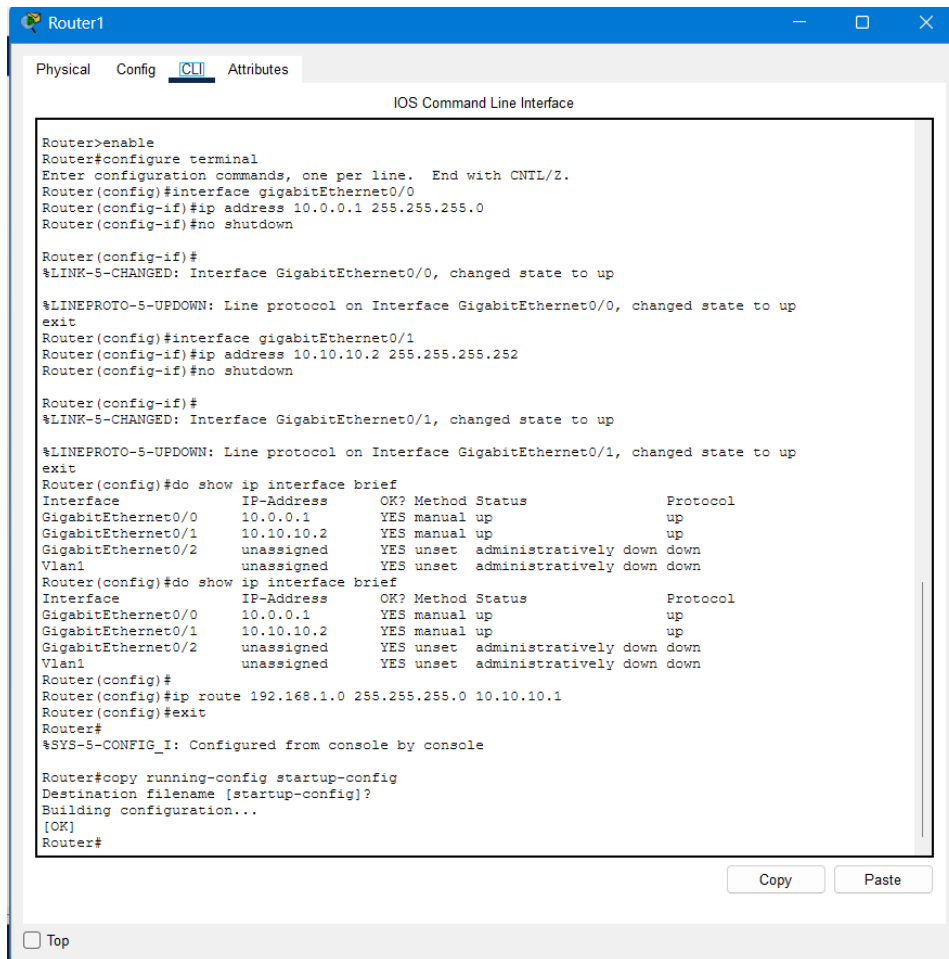
```

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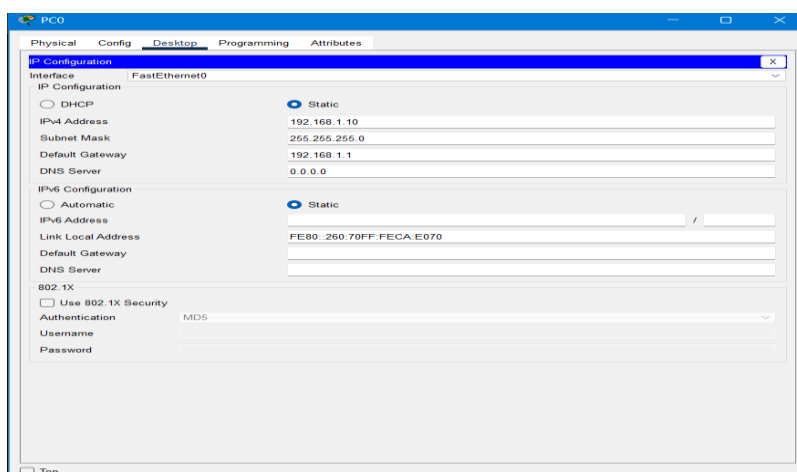
Router0 interface configuration.

Router1 Configuration:



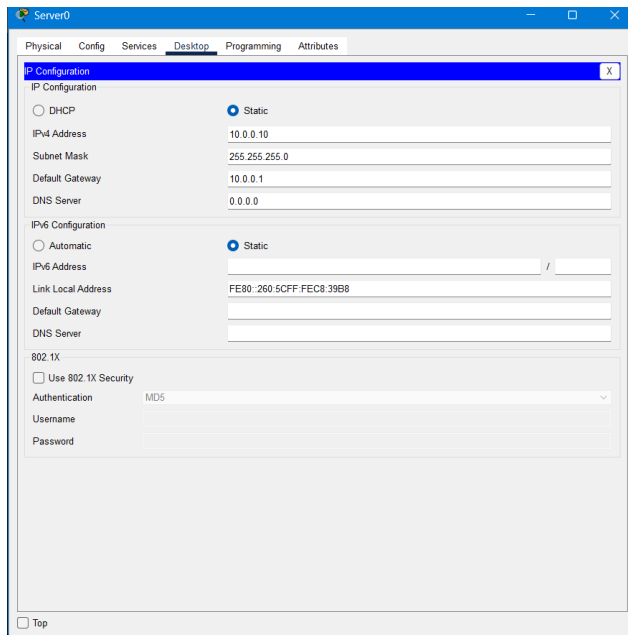
Router1 interface configuration.

PC Configuration:



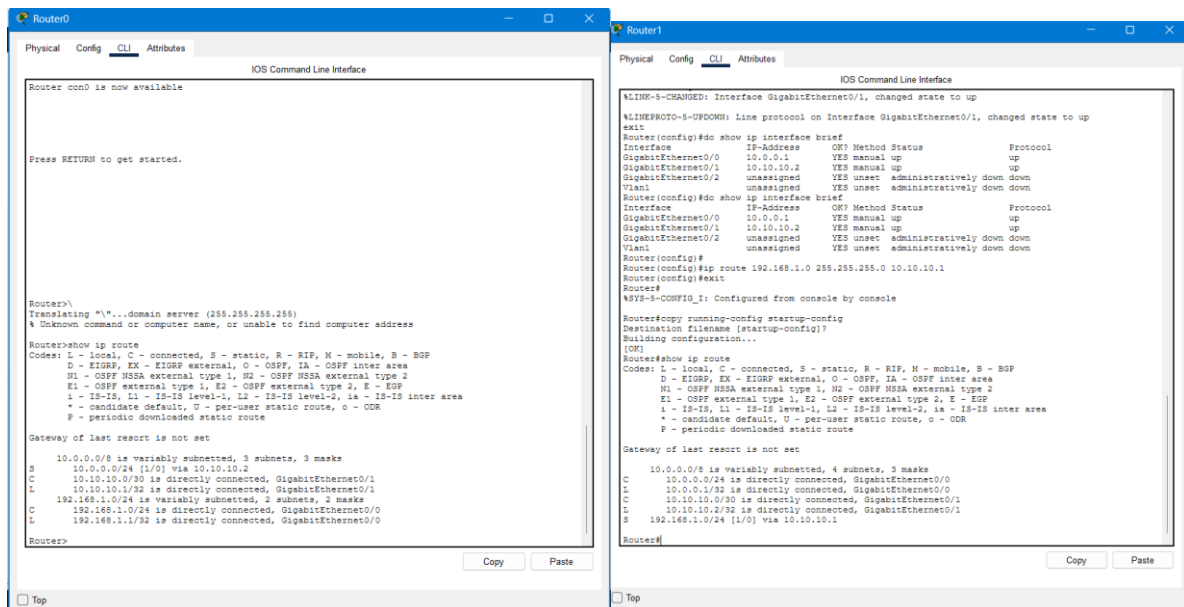
PC IP configuration.

Server Configuration:



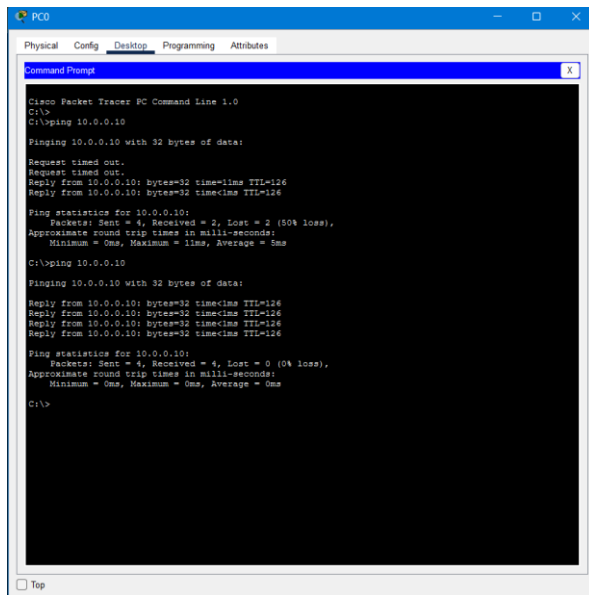
Server IP configuration.

Static Route Configuration:



Static routing configuration.

Connectivity Test:



```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 5ms

C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Successful ping from the PC to the cloud server.

Result:

The hybrid cloud network was successfully simulated using Cisco Packet Tracer. Two routers were configured with IP addressing and static routing to interconnect the on-premises network and the cloud network. End-to-end communication between the PC and the server was verified successfully using the ping command, confirming successful hybrid cloud connectivity.